10.5 A 12b 600MS/s Pipelined SAR and 2×-Interleaved Incremental Delta-Sigma ADC with Source-Follower-Based Residue-Transfer Scheme in 7nm FinFET

Seungyeob Baek, Ilhoon Jang, Michael Choi, Hyungdong Roh, Woongtaek Lim, Youngjae Cho, Jongshin Shin

Samsung Electronics, Hwasung, Korea

2021 IEEE International Solid- State Circuits Conference (ISSCC) | 978-1-7281-9549-0/20/$31.00 ©2021 IEEE | DOI: 10.1109/ISSCC42613.2021.9366051

Recently, pipelined ADCs are frequently adopted as sub-ADCs in multi-channel time-interleaved ADCs because a pipelined manner can reduce the number of interleaved channels by maximizing single-channel ADC operating speed. To reduce power consumption of residue amplifier, previous works [2,4,5] have explored power efficient residue amplifier architecture, such as dynamic amplifiers (DAs), open-loop amplifiers (OAs) and ring amplifiers (RAs). However, inter-stage gain error of power-efficient DAs, OAs, and RAs needs power-consuming gain calibration to compensate process, power supply, and temperature (PVT) variations.

This work presents a gain-calibration-free pipelined ADC architecture with the use of an open-loop residue-transferring buffer. The ADC consists of the 1st-stage successive-approximation (SAR) ADC, the 2nd-stage delta-sigma modulator (DSM), and a residue buffer. Instead of amplifying residue signal, the residue buffer transfers residue signal to the 2nd stage for pipelined operation. The non-amplified small residue signal is resolved by the 2nd stage DSM, which has wide dynamic range. A capacitor DAC (CDAC) in the 1st stage and a residue buffer are replicated in the feedback path of the 2nd stage DSM to eliminate PVT-variation-oriented inter-stage gain error without gain calibration. In addition, duration-fixed synchronous-like asynchronous clock, adaptive speed-controlled (ASC) clock, guarantees robust operation of the SAR against PVT variations. At 300MS/s, the gain-calibration-free ADC achieves a peak SNDR of 58.2dB. Moreover, the ADC has a 55.3dB SNDR with a 1Vpp swing, 500MHz (beyond Nyquist) input at 600MS/s. The ADC has superior speed over a wide input bandwidth compared with a sub-ranging SAR/DSM [1] and a noise-shaping pipelined SAR [3].

The ADC architecture and the corresponding timing diagram are shown in Fig. 10.5.1. The proposed ADC consists of a 7b single-channel SAR 1st stage, a 6.5b 2-channel time-interleaved 2nd-order continuous-time (CT) incremental DSM 2nd stage, and a residue buffer. The 1.5b redundancy can relax an inter-stage offset error requirement. The 1st stage samples the input signal with bootstrapped bottom-plate sampling. The sampling capacitance is 690fF to meet kT/C noise and residue signal accuracy. In the conversion phase, the 1st stage SAR resolves 7 bits with equally periodic 7 cycles of internal operation clock, CK_{ASC}, which triggers the comparator and SAR logic. After 7b conversion, the residue signal in a CDAC of the 1st stage is transferred to the 2nd stage by an open-loop residue buffer. The gain error of the residue signal path causes inter-stage gain error, which comes from the parasitic capacitance of a CDAC in the 1st stage and a gain error of open-loop residue buffer. The prototype ADC eliminates inter-stage gain error by matching the 2nd stage feedback path gain to residue signal path gain, rather than adopting digital calibration. The 2nd-stage CT incremental DSM operates at 300MS/s with 14 cycles of a 4.8GHz operation clock, CK_{DSMC}, which is buffered and enveloped from an externally applied CK_M. The 2-channel DSM operates in a time-interleaved manner to achieve a 600MS/s sampling rate. The linearity requirement of the time-interleaved 2nd stage is relaxed because the 2nd stage samples timing skew-less residue signal. As a result, the 2nd stage requires only offset calibration between interleaved ADCs.

The operating principle of the proposed architecture is described by the equation in Fig. 10.5.2. Quantization noise (Q-noise) consists of a gain mismatch term and a noise-shaping term. The noise-shaping term can be minimized under sufficiently high loop gain, H(s). Also, the gain mismatch term can be minimized by eliminating gain mismatch between the residue signal path and the feedback path of the 2nd stage. Fig. 10.5.2 shows the circuit implementation of residue buffer and the 2nd stage single-channel DSM. The 2nd-order noise-shaping DSM with 14 cycles of conversion clock is a proper choice in the 2nd stage to increase H(s), and it results in 6.5b resolution including 1.5b redundancy. Although the Gm-C integrator consumes less power, an active-RC integrator is widely used in CT DSM because the Gm-C integrator has a narrow linear input range. However, the coarse-converted residue signal enables the DSM to apply a power-efficient Gm-C integrator. A simple open-loop source follower is applied as a residue buffer. Since the source follower is robust to PVT variations and easy to implement, the design complexity can be relaxed. A CDAC and a residue buffer are replicated in the 2nd stage in Fig. 10.5.2 to eliminate the inter-stage gain error. For better gain matching, a comparator of the 1st stage is also replicated in the feedback path as a dummy. Well-matched replica circuits with shared-bias and adjacent layout arrangements remove inter-stage gain error from chip-to chip gain variation of the residue signal path. Moreover, the area-consuming 7b CDAC can be used in calibrating inter-stage offset error by adding offset in the feedback signal. Therefore, the comparator without offset calibration enables the SAR to enhance conversion speed.

Figure 10.5.3 shows the proposed adaptive speed-controlled (ASC) clock generation scheme. The ASC clock generator takes advantages from both the synchronous and asynchronous schemes by generating a 50% duty and duration-fixed internal operation clock (CK_{ASC}) cycles, regardless of the PVT variations, by background frequency calibration. The ASC clock generator dedicates at least a 1/2 clock period of CK_{ASC} for the internal DAC settling time. In fact, the DAC settling initiates as soon as the comparator makes the decision (just like the asynchronous operation) that dedicates more than ½ the clock period of CK_{ASC} for additional performance enhancement. At the end of the data conversion, a shift-register in the SAR logic counts the number of CK_{ASC} cycles and generates the end-of-conversion (EOC) signals. As shown in Fig. 10.5.3, the ASC clock generator calibrates the frequency of CK_{ASC} in the background by reducing or increasing the delay of the clock generator. The ASC clock generator consists of an inverter and capacitor-based ring oscillator and the delay line controller. Since the input sampling is performed by low-jitter CK_{SAMP}, the delay-controlled ring oscillator only cares to generate seven CK_{ASC} cycles with a relaxed jitter requirement for SAR operation. In addition, the duration-fixed synchronous-like asynchronous ASC clock helps the ADC easily detect the comparator meta-stability like a synchronous SAR.

The prototype ADC, implemented in 7nm FinFET CMOS technology, occupies 0.037mm^2 as shown in Fig. 10.5.7. The foreground on-chip offset calibration is performed to match the signal range between the 1st and the 2nd stage. Moreover, off-chip offset calibration is done between the 2-channel DSMs. No inter-stage gain calibration is applied with the help of the feedback path replica circuits. At 600MS/s under a 0.8V supply, the total power consumption is 13mW. The residue source follower consumes only 0.7mW. Measured performance is shown in Fig. 10.5.4. The ADC operates over a wide range (300 ~ 600MS/s) of sampling rates, although it adopts a CT DSM. The ADC achieves a peak SNDR of 58.2dB and a peak SFDR of 79.4dB at 300MS/s with a 5MHz input. At 600MS/s, the peak SNDR is 56.6dB, and it degrades less than 1.3dB up to 500MHz input. The measured SNDR at 600MS/s stays above 54dB across a 0.77~0.9V power supply range and -40~80°C temperature variations. Figure 10.5.5 illustrates the measured spectra and DNL/INL. The achieved DNL and INL are 0.71 and 1.45LSB, respectively. Figure 10.5.6 compares the ADC performance with the state-of-the-art ADCs. The proposed gain calibration-free ADC shows outstanding performance beyond the Nyquist frequency with only 1Vpp input swing.

References:

[1] A. Venca et al., "27.8 A 0.076mm2 12b 26.5mW 600MS/s 4×-interleaved subranging SAR-ΔΣ ADC with on-chip buffer in 28nm CMOS," *ISSCC*, pp. 470-472, Feb. 2016.
[2] B. Hershberg et al., "3.6 A 6-to-600MS/s Fully Dynamic Ringamp Pipelined ADC with Asynchronous Event-Driven Clocking in 16nm," *ISSCC*, pp. 68-70, Feb. 2019.
[3] Y. Song et al., "9.6 A 2.56mW 40MHz-Bandwidth 75dB-SNDR Partial-Interleaving SAR-Assisted NS Pipeline ADC With Background Inter-Stage Offset Calibration," *ISSCC*, pp. 164-166, Feb. 2020.
[4] J. Lagos et al., "A single-channel, 600Msps, 12bit, ringamp-based pipelined ADC in 28nm CMOS," *IEEE Symp. VLSI Circuits*, pp. C96-C97, June 2017.
[5] K. Moon et al., "A 9.1 ENOB 21.7fJ/conversion-step 10b 500MS/s single-channel pipelined SAR ADC with a current-mode fine ADC in 28nm CMOS," *IEEE Symp. VLSI Circuits*, pp. C94-C95, June 2017.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on May 29,2026 at 05:25:45 UTC from IEEE Xplore. Restrictions apply.

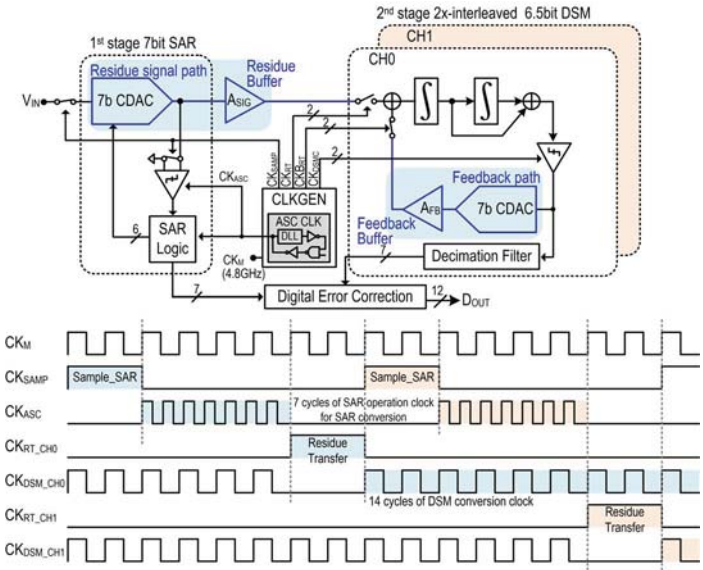


Figure 10.5.1: Block and timing diagram of the prototype 12b pipelined SAR and 2×-interleaved CT incremental DSM.

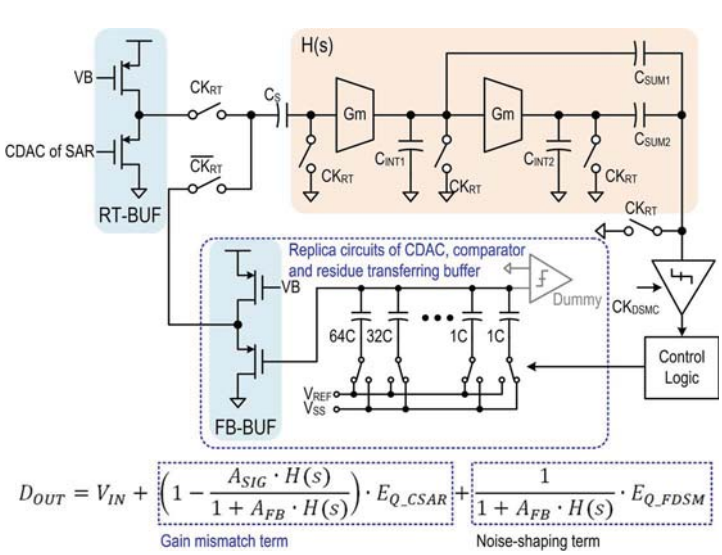


Figure 10.5.2: The circuit implementation of the 2nd stage and the operation principle of gain-error-calibration-free residue transfer scheme.

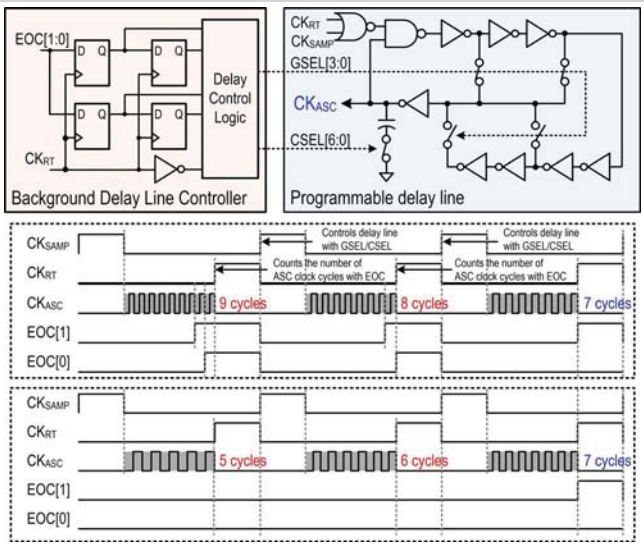


Figure 10.5.3: The circuit implementation of the adaptive speed-controlled (ASC) clock generator in the 1st stage (Top) and timing diagram of background calibration by increasing (middle) or reducing (bottom) delay.

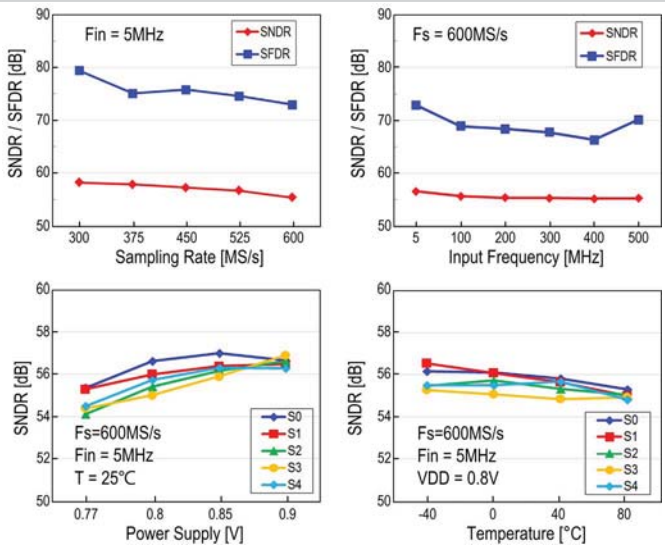


Figure 10.5.4: Dynamic performance and measured SNDR across power supply (0.77V ~ 0.9V) and temperature (-40 to 80°C) variations of 5 randomly selected chips at 600MS/s.

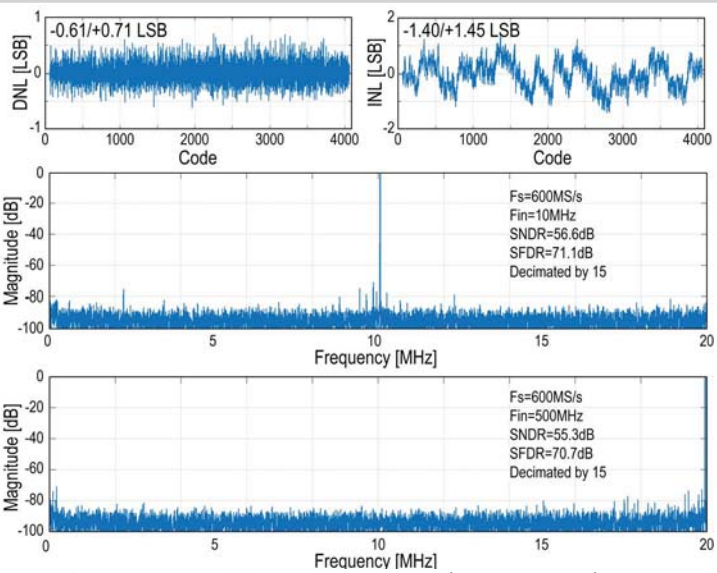


Figure 10.5.5: Measured 16384-point output spectra (decimated by 15) and DNL/INL at 600MS/s.

	This work	[1] ISSCC2016	[2] ISSCC2019	[4] VLSI2017	[5] VLSI2017
Architecture	**Pipe-SAR &TI DSM**	TI subranging SAR&DSM	Pipeline	Pipeline	Pipe-SAR
Technology	**7nm**	28nm	16nm	28nm	28nm
Resolution [bits]	**12**	12	11	12	10
Sample Rate [MS/s]	**600**	600	600	600	500
Supply Voltage [V]	**0.8**	1.2/1.5	0.85	0.9	1
Input swing [$V_{pp,diff}$]	**1**	1.5	1.5	1.6	1.6
Input Bandwidth [MHz]	**500**	265	600		250
SFDR @ Nyq. [dB]	**67.8**	-	78.3	69.2	69.2
SNDR @ Nyq. [dB]	**55.3**	58	60.2	56.3	56.6
Power [mW]	**13**	26.5	6	14.2	6
FoM_{Walden} @ Nyq. [fJ/c-s]	**45.6**	68	12	44.3	21.7
$FoM_{Schreier}$ @ Nyq. [dB]	**158.9**	158	167.2	159.5	162.8
Area [mm²]	**0.037**	0.076	0.037	0.62	0.015
Gain Calibration	**X**	O	O	O	O

Figure 10.5.6: Comparison with state-of-the-art.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on May 29,2026 at 05:25:45 UTC from IEEE Xplore. Restrictions apply.

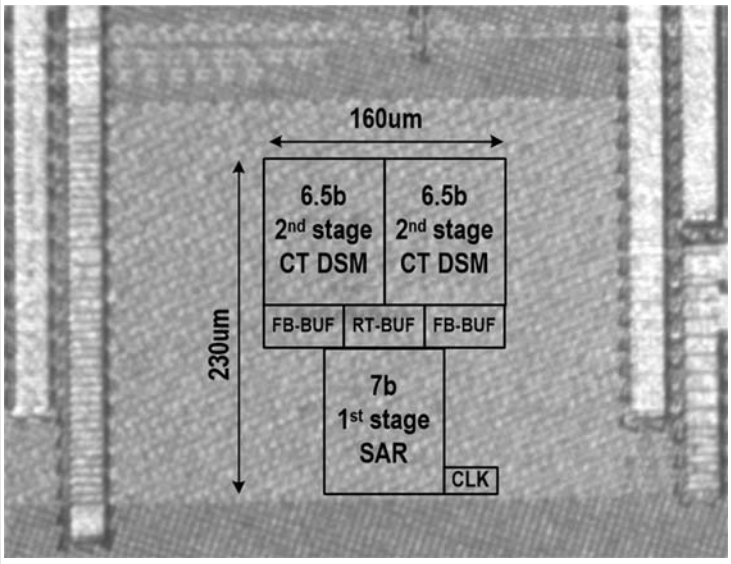


Figure 10.5.7: Die photo.

Authorized licensed use limited to: Tamkang Univ.. Downloaded on May 29,2026 at 05:25:45 UTC from IEEE Xplore. Restrictions apply.

978-1-7281-9549-0/21/$31.00 ©2021 IEEE